IOT & EMBEDDED SYSTEMS

RFID Smart Access & Attendance System

ESP8266 • RC522 • LCD 16X2 • BUZZER • LEDS • JUMPER WIRES

Presented by Alok Ghewande

Project Overview

System Foundations

Combining wireless frequency identification, microcontrollers, and multi-modal feedback for next-gen physical security.

Problem Statement & Objectives

The Problem

- **Manual Logging Issues:** Paper-based attendance registries are slow, vulnerable to proxy entries, and prone to human error.
- **Lack of Real-Time Audit:** Traditional mechanical keys offer zero digital audit trail or real-time entry monitoring.
- **No Visual/Audio Status:** Basic RFID readers lack clear onboard visual status displays and audio confirmations for users.

Project Objectives

- **Instant UID Authentication:** Securely read and parse 13.56 MHz RFID tags using the MFRC522 module under 300ms.
- **Clear Character Display:** Provide real-time operational feedback using an LCD 16x2 display (via I2C interface).
- **Audio-Visual Feedback:** Trigger instant Green/Red LED indicators and active piezo buzzer beeps upon scanning.

Core Hardware Architecture Stack

ESP8266 NodeMCU

Main 32-bit Wi-Fi MCU driving system logic, processing tag UIDs, and enabling cloud logging capability.

RC522 RFID Reader

13.56 MHz High-frequency contactless transceiver module communicating via SPI protocol.

LCD 16x2 Display

Alphanumeric display with I2C backpack for real-time status alerts ("Access Granted / Denied").

Active Buzzer

Provides distinct frequency tone alerts for successful scans vs unauthorized intrusion attempts.

Status LEDs

Dual LED indicators (Green for Granted, Red for Denied) giving instant visual status at a glance.

Jumper Wires & Rails

High-grade Dupont Male-to-Female/Male-to-Male jumper wires ensuring stable SPI, I2C, and GPIO links.

Microcontroller: ESP8266 NodeMCU

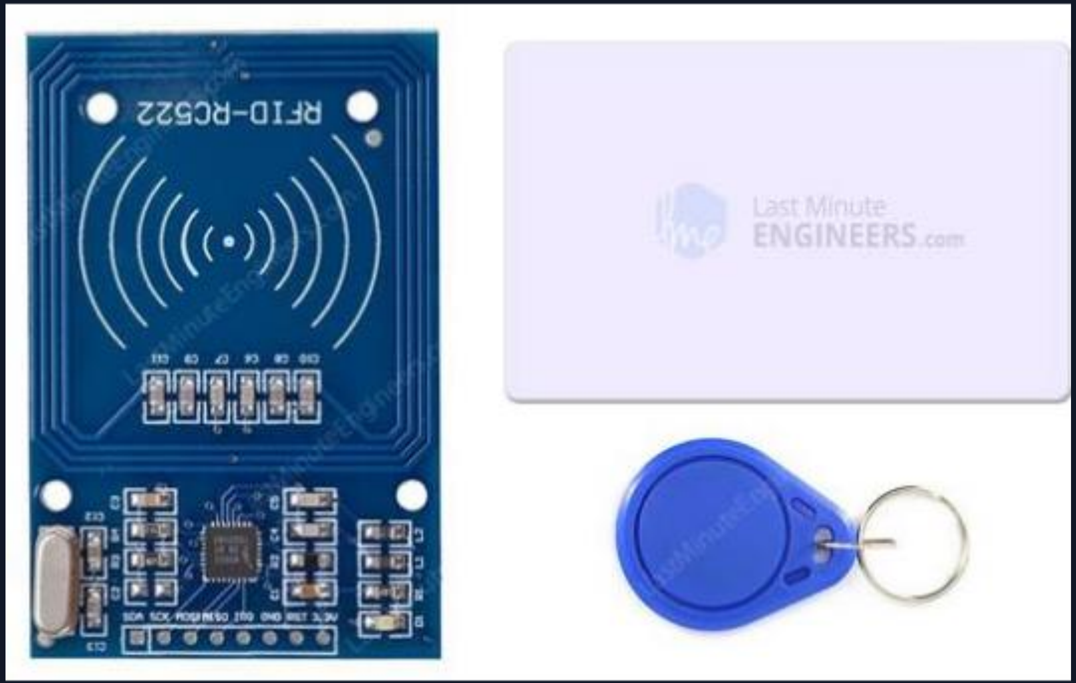
The **ESP8266 NodeMCU** serves as the brain of the access control system, managing peripheral communication and Wi-Fi network telemetry.

- **Processor Core:** Tensilica L106 32-bit RISC microprocessor running at 80/160 MHz.
- **Wireless Networking:** Integrated 802.11 b/g/n Wi-Fi transceiver for IoT cloud syncing.
- **Peripheral Support:** Hardware SPI for RC522, I2C for LCD 16x2, and general GPIOs for signaling.
- **Operating Voltage:** 3.3V power domain with built-in USB power regulator.

RFID Sensing: MFRC522 Transceiver

The **MFRC522** is a highly integrated contactless reader/writer IC operating on the 13.56 MHz communication band.

- **Operating Frequency:** 13.56 MHz ISO/IEC 14443 A / MIFARE standard.
- **Host Interface:** SPI (Serial Peripheral Interface) for high-speed data transfer to ESP8266.
- **Read Range:** Up to 50mm distance, ideal for quick contactless card swipes.
- **UID Parsing:** Extracts unique 4-byte or 7-byte factory hardware identification codes.



Visual Interface: LCD 16x2 Module

The **LCD 16x2 character display** provides explicit user feedback at the entryway, displaying welcome messages or access rejection warnings.

- **Display Capacity:** 16 characters across 2 rows with adjustable LED backlighting.
- **I2C Serial Adapter:** Utilizes PCF8574 I2C expansion module to reduce pin usage from 6 GPIOs down to just 2 lines (SDA & SCL).
- **Dynamic Text Refresh:** Instantly displays user name upon authorization or "Access Denied" for unknown UIDs.
- **Low Power Consumption:** Consumes minimal current, making it ideal for continuous wall-mounted entry stations.

16x2 Character LCD Display with Blue Backlight

Hardware Interfacing

Jumper Wires & Circuit Assembly

Reliable hardware connections are critical for high-speed SPI signal integrity and low noise. High-grade Dupont jumper wires link the ESP8266 controller to the RC522 RFID reader, LCD 16x2 display, active buzzer, and LED indicators.

Common ground planes and regulated 3.3V/5V rails prevent voltage spikes and signal drops during active RFID card detection and buzzer sound triggering.

Pin Interfacing & Mapping Table

Peripheral Component	Module Pin	ESP8266 GPIO Pin	Communication Protocol / Type
RC522 RFID Reader	SDA (SS) / SCK / MOSI / MISO	GPIO15 (D8) / GPIO14 (D5) / GPIO13 (D7) / GPIO12 (D6)	SPI Protocol SPI Protocol
RC522 RFID Reader	RST (Reset) / GND / 3.3V	GPIO0 (D3) / GND / 3.3V Pin	Power & Control
LCD 16x2 Display (I2C)	SDA / SCL / VCC / GND	GPIO4 (D2) / GPIO5 (D1) / 5V / GND	I2C (0x27 / 0x3F) I2C (0x27 / 0x3F)
Active Buzzer	Positive (+) Pin / GND	GPIO2 (D4) / GND	Digital Output (PWM Tone)
Green & Red LEDs	Anodes (+) via 220Ω / GND	GPIO16 (D0) & GPIO0 (D3) / GND	Digital GPIO Output

System Operational Sequence

1. Card Swipe

User taps 13.56 MHz card near RC522. RFID reader energizes tag and extracts unique hardware UID.

2. UID Verification

ESP8266 receives UID via SPI and checks against authorized memory database or database server.

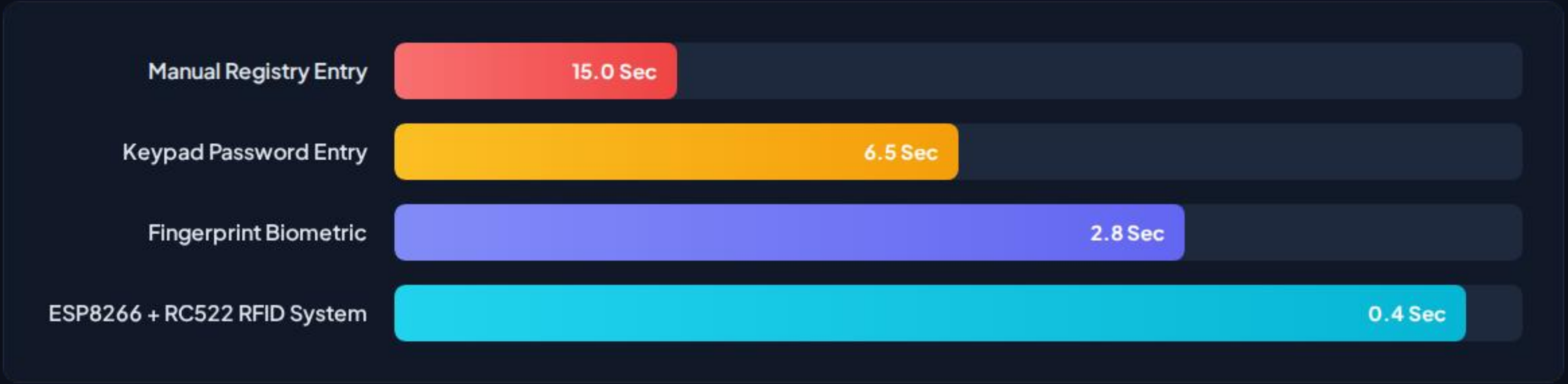
3. LCD Refresh

LCD 16x2 via I2C displays "Access Granted: Name" or "Access Denied: Invalid Card".

4. Audio-Visual Alert

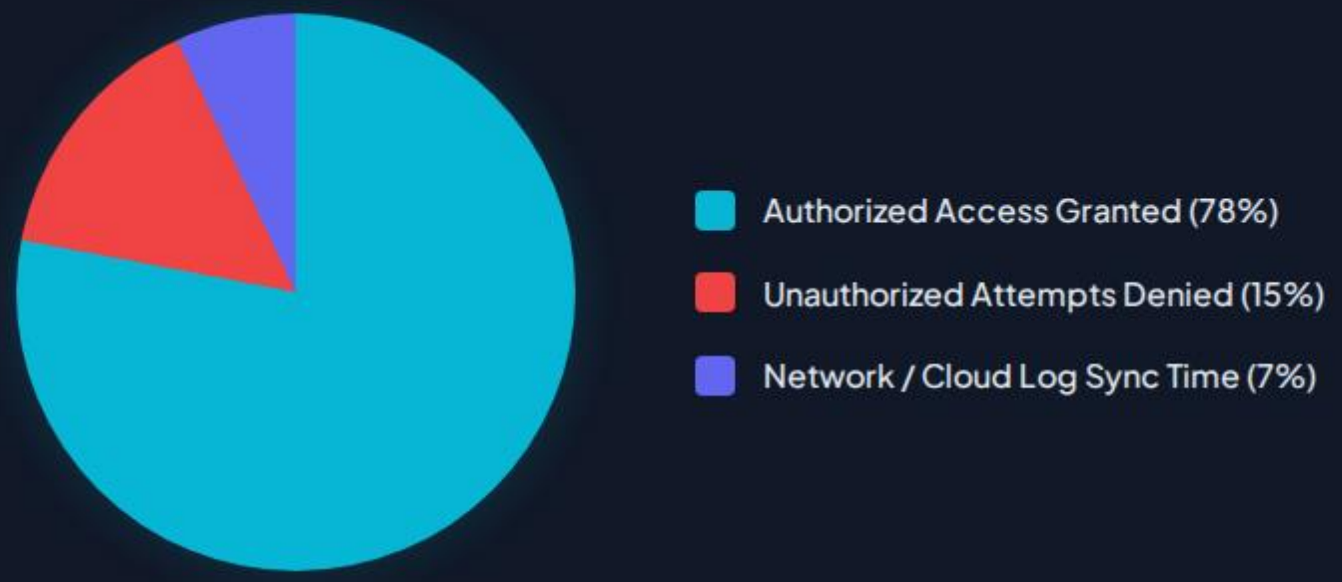
ESP8266 triggers Green LED + single short beep for authorized entry, or Red LED + long alarm beep.

System Speed & Efficiency Metrics



The RFID + ESP8266 architecture achieves sub-second contactless verification (0.4s average authentication delay), drastically reducing entryway queues.

Access Verification Distribution



Real-world operational telemetry showing fast local decision making with background cloud attendance logging over Wi-Fi.

Key Performance Highlights

<0.5s

Total Scan-to-Alert Latency

Reliable & Low-Cost Embedded Architecture

By leveraging the ESP8266's fast hardware SPI and hardware I2C modules alongside dedicated digital interrupt pins, the system guarantees zero user lag during peak access hours.

Dual feedback via the 16x2 LCD display and active buzzer ensures zero ambiguity for users attempting entry.

Real-World Deployment Use Cases

Corporate Office Entry

Automated staff clock-in/clock-out tracking with instant digital logging and monthly attendance reporting.

Research Lab Security

Restricted access control allowing only authorized personnel into high-containment chemical or server rooms.

School & University Halls

Classroom and examination hall attendance tracking, replacing manual roll calls with contactless RFID tapping.

Questions & Discussion

Thank you for your time and attention!

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Image Sources

https://www.make-it.ca/wp-content/uploads/2021/09/nodemcu-esp8266-small-162x300-1.jpg

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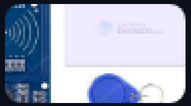

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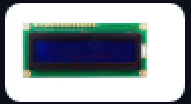

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